2.2 Characteristics of Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.2 · Student sheet

I can…

- find the vertex, axis of symmetry, and y-intercept of a quadratic in standard form and graph it.
- find the minimum or maximum value and the domain and range.
- use intercept form to find the x-intercepts and vertex.

Vocabulary · fill in the missing word

Axis of symmetry — the _____ line x = h that divides a parabola into mirror images

Standard form — $f(x) = ax^2 + bx + c$; the axis is $x = -b/(2a)$ and the _____ is c

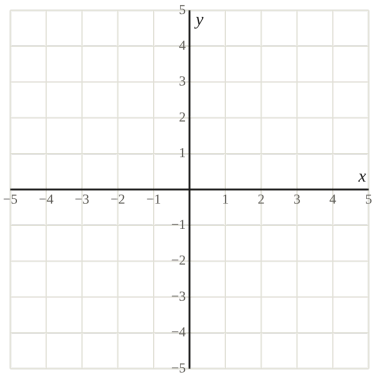
Minimum / maximum value — the _____ of the vertex when $a > 0$ / $a < 0$

Intercept form — $f(x) = a(x - p)(x - q)$; the _____ are p and q

Part 1 · Fold the parabola

x	0	1	2	3	4
$y = x^2 - 4x + 3$					

Plot the points, sketch the parabola, and fold along the line through the lowest point.



Axis of symmetry: _____ Vertex: _____ Which points matched? _____

Part 2 · Notes: fill in the blanks

Standard form: the axis of symmetry is x = _____, and the y-intercept is _____.

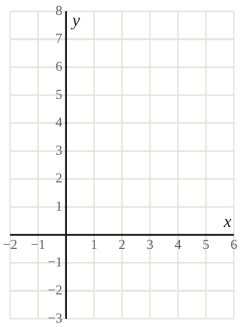
Intercept form $a(x - p)(x - q)$: the x-intercepts are _____.

$a > 0$: _____ value; $a < 0$: _____ value.

Part 3 · Examples (work them with the class)

Example 1

Graph $f(x) = 2x^2 - 8x + 6$. Label the vertex and the axis of symmetry. Give the minimum value and the range.



Example 2

Find the minimum or maximum value of $f(x) = -x^2 - 6x + 4$. Give the domain and range.

Example 3

Find the x-intercepts, the axis of symmetry, and the vertex of $f(x) = -2(x + 3)(x - 1)$.

Example 4

A store's weekly profit P (in dollars) from an item is $P = -2x^2 + 40x - 50$, where x is the price (in dollars). What price gives the maximum profit? What is the maximum profit?

Part 4 · Partner practice

Partner 1

Find the vertex and the axis of symmetry of $f(x) = x^2 + 4x - 1$.

Partner 2

Find the x-intercepts, the axis of symmetry, and the vertex of $f(x) = (x - 2)(x - 6)$.